



From breaking news to approved mitigation to drafting customer emails - counteracting supply chain disruptions for SMEs/startups

The Problem

Supply-chain shocks arrive as fragmented news, advisories and logistics updates. SMEs often discover the signal late, verify it manually, and then struggle to connect it to hidden dependencies across suppliers, routes, inventory, schedules and customer orders.



One external event can propagate through ports, fabs, components, production lines and customer orders

Why Chainsilience AI

Company-specific

Maps verified events to the exact supplier, site, component, route, etc.

Quantitatively grounded

LLMs handle language; deterministic calculators, simulation and optimization handle numbers.

Evidence-preserving

Every alert retains sources, confidence, graph paths, assumptions, twin version and approval history.

Governed action

The system proposes and drafts; an authorized human approves the exact mitigation/ communication.

Value Exchange

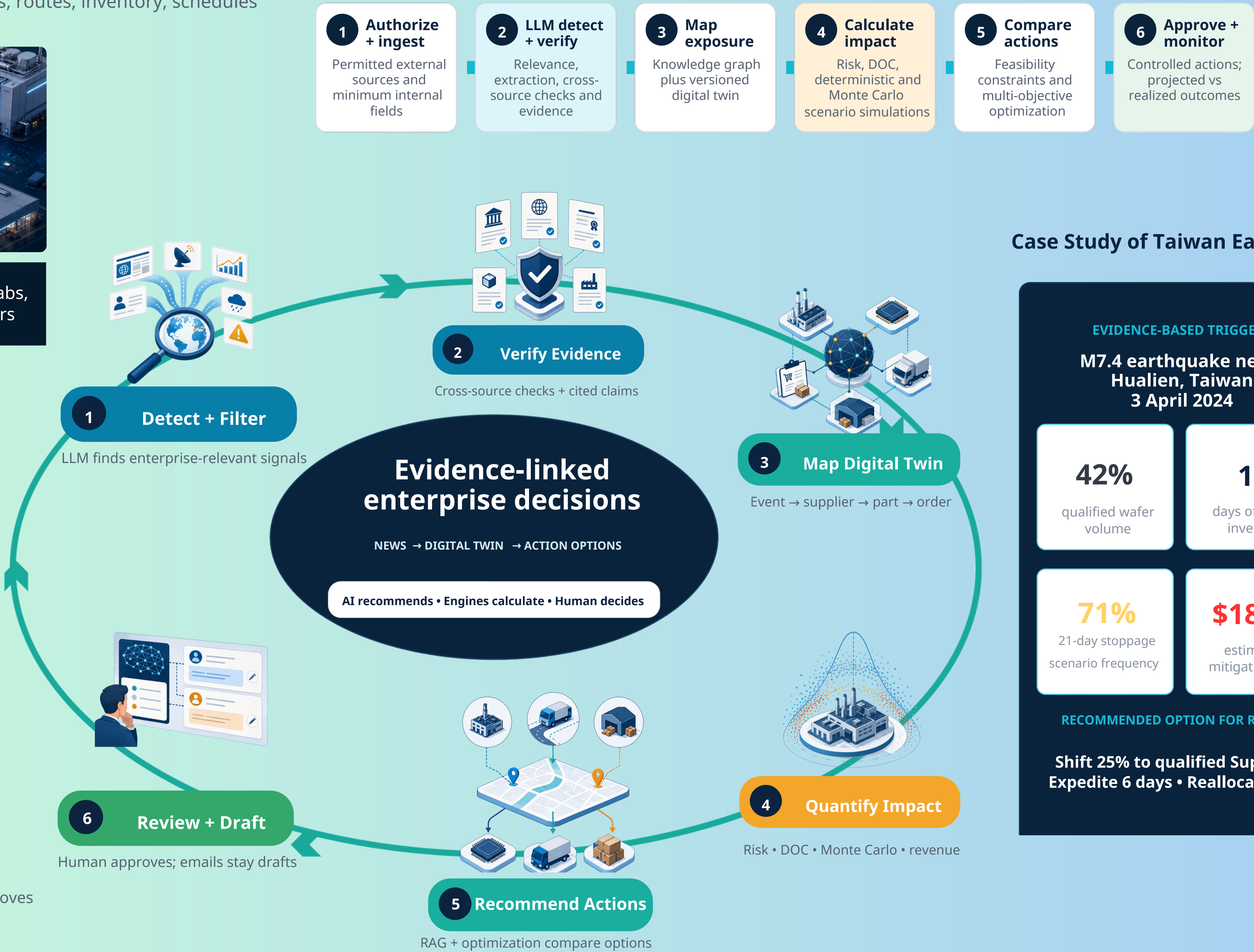
The company contributes authorized operational data, constraints and decision rules.

Chainsilience AI returns earlier warning, quantified exposure, feasible response options and auditable communication; the company retains decision authority. Value is measured through avoided expected loss, analyst time saved and improved service continuity relative to operating cost.

Start narrow: one product family, a limited supplier set and a small number of trusted sources.

End-to-End Governed Workflow

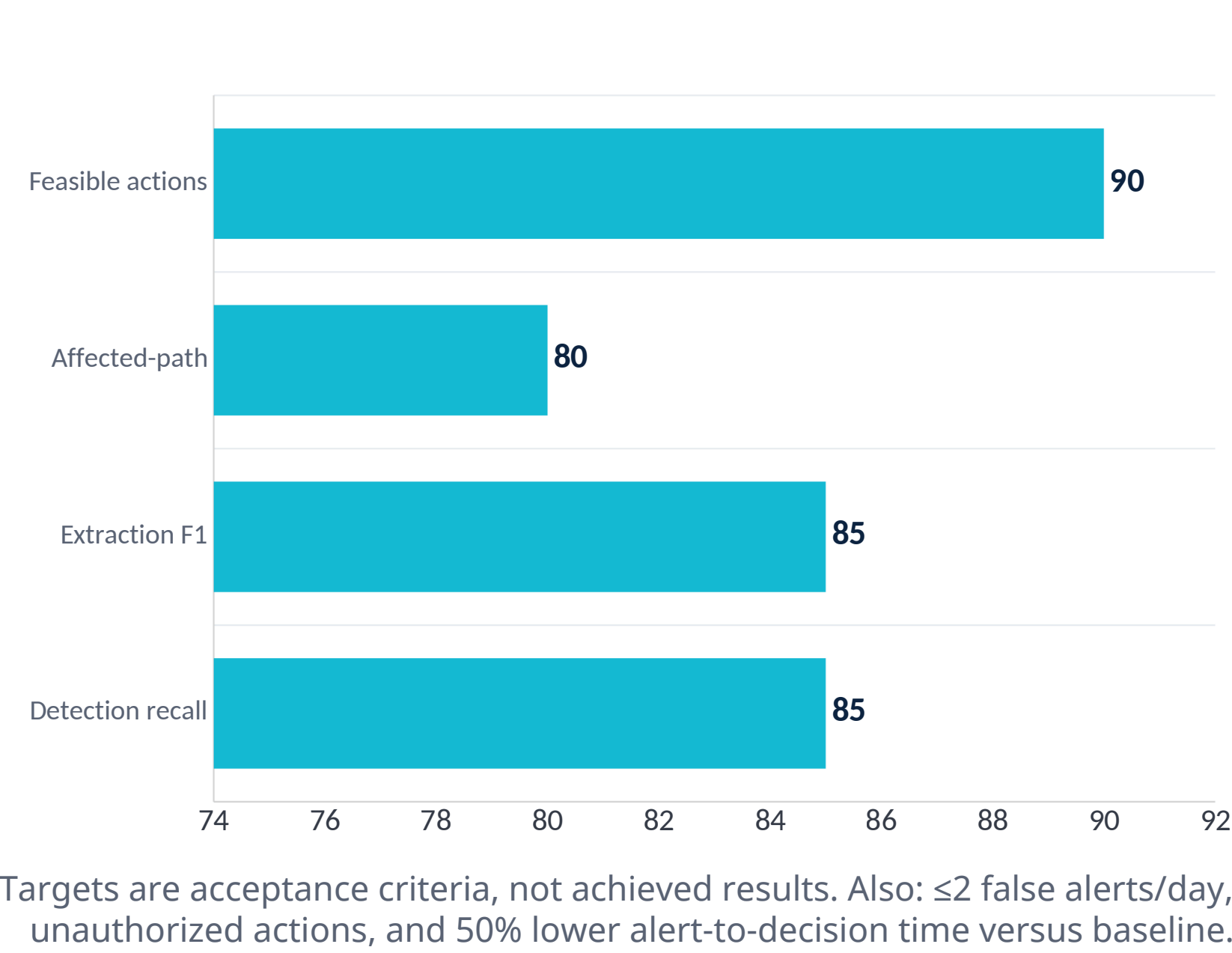
The LLM understands language, relevance, and context; analytical engines calculate impact; company rules enforce feasibility; an authorized human decides.



Case Study of Taiwan Earthquake



Pilot Acceptance Criteria



Governed Agentic AI

- Encryption, role/field access and tenant isolation
- Policy gateway; no unauthorized sends or record changes
- Human approval of exact strategy, budget and message
- Baseline, projected and realized outcomes kept separate

LLM: understand, coordinate, explain and draft.
Engines: calculate. Human: decide.

GLOSSARY + KEY CONTROLS

Agentic AI

Specialized agents use authorized tools to retrieve, structure, analyse and draft outputs. Consequential actions remain policy-controlled and reviewable.

LLM Relevance + Verification

The LLM identifies company-relevant news, extracts event facts, compares sources, detects contradictions and flags unsupported or low-confidence claims.

Knowledge Graph

A network connecting events to suppliers, sites, components, routes, products, inventory and customer orders through explicit relationships.

Digital Twin

A versioned, time-indexed computational replica of the supply network and its operational state for reproducible what-if scenarios.

Retrieval-Augmented Generation (RAG)

Generation grounded in retrieved records and evidence passages, reducing unsupported summaries and keeping claims traceable.

Monte Carlo Simulation

A numerical engine randomly samples uncertain disruption duration, recovery, demand, yield and capacity to estimate scenario frequency. The LLM does not perform the calculation.

Model Context Protocol (MCP)

A possible interoperability layer for exposing typed, authorized resources and tools. Authentication, authorization and approval remain application responsibilities.

Natural Language Processing (NLP)

Technology for analyzing, and generating human language. NLP is used here for relevance filtering, event extraction, risk summaries, and context-aware email drafts.

Internal Data Confidentiality

Sensitive supplier, route, product, order, inventory and schedule data must be encrypted and restricted by strict access controls. It must not be disclosed to third parties or used to train external AI models without explicit company consent.

Human approval supports accountability, but does not guarantee ethical or correct outcomes.